

VideoRoPE: What Makes for Good Video Rotary Position Embedding?

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Outline

- Motivation and problem framing
- Method: VideoRoPE (LTA, DL, ATS)
- Diagnostics: V-NIAH-D and attention analyses
- Results: understanding, retrieval, hallucination, extrapolation
- Ablation insights and takeaways

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Background: Video LLMs

- Long-form video understanding powers search, assistants, education, and safety.
- Transformers need positional encodings to reason over order and duration.
- Scaling to hour-long videos demands accurate spatio-temporal grounding.

Problem: Video Position Encoding

- 1D RoPE excels in text but ignores video's 3D structure (t, x, y).
- Naïve extensions mis-handle temporal distances and spatial symmetry.
- Outcome: drift, aliasing, and brittle retrieval over long videos.

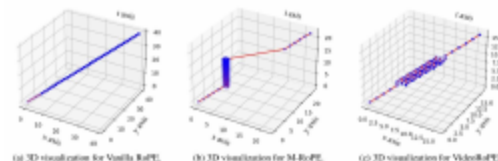


Figure: 3D位置编码对比: 1DRoPE无空间; M-RoPE跨帧索引不一致; VideoRoPE兼顾空间与一致增长

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- RoPE frequency depends on dimension index—axis assignment changes periodicity.
- High-frequency temporal encoding causes oscillations and collisions.
- Spatial and temporal indices must be decoupled yet remain coherent.

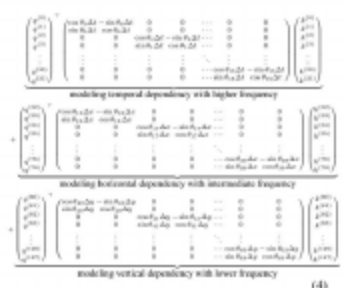


Figure: 频率分配：在d维中分配t/x/y, 3D结构下策略不等价

- Design a 3D RoPE that preserves spatial symmetry and long-range temporal order.
- Assign temporal to low frequencies; interleave x/y; keep text–video continuity.
- Introduce adjustable temporal spacing to align text and video granularities.

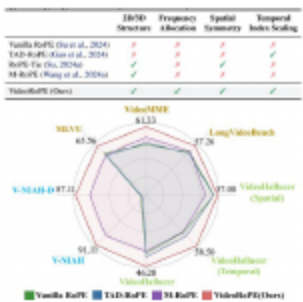


Figure: VideoRoPE outperforms RoPE variants on benchmarks.

Contributions at a Glance

- Four criteria for video RoPE: 3D structure, frequency allocation, spatial symmetry, temporal scaling.
- VideoRoPE with LTA, DL, ATS; plug-in replacement in Video LLMs.
- V-NIAH-D: hard retrieval with periodic distractors to stress-test time encoding.

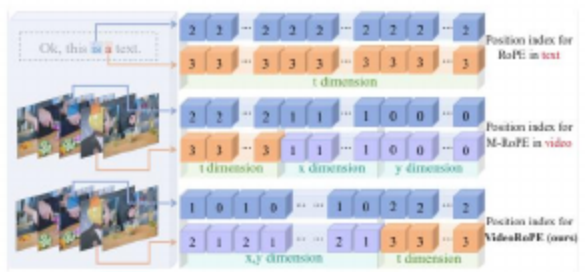


Figure: Vanilla RoPE、M-RoPE与含交错时空设计的VideoRoPE在相邻token/帧的嵌入对比

Method: VideoRoPE Overview

- Compute attention with relative rotations over temporal and spatial offsets.
- Interleaved x/y occupy lower dims; t placed in higher, low-frequency dims.
- Maintain text-like index growth and coherent video token geometry.

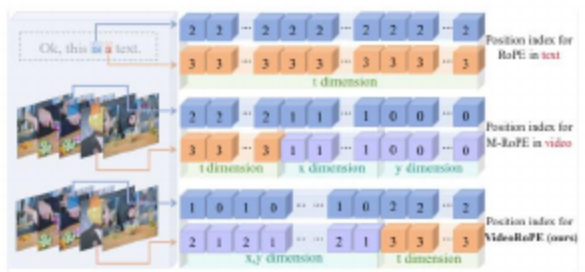


Figure: Vanilla RoPE、M-RoPE与含交错时空设计的VideoRoPE在相邻token/帧的嵌入对比

- Interleave x/y in low dims to balance spatial modeling.
- Allocate temporal to late rotary angles for global time capacity.
- Dimension counts (for fairness): x=48, y=48, t=32 (matching M-RoPE).

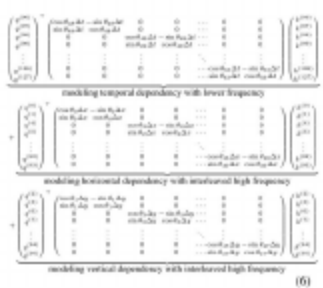


Figure: The horizontal position x and vertical position y are inter-

- Map time to lowest-frequency angles for wide monotonic regions.
- Prevents oscillatory wrap-around and temporal collisions.
- Enables robust long-range temporal discrimination.

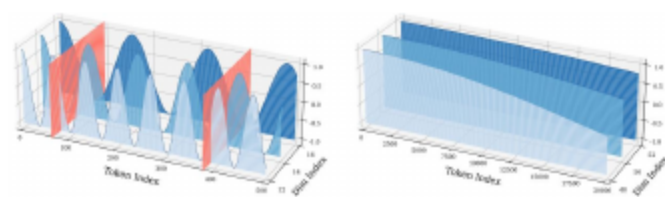


Figure: (a) Temporal Frequency Allocation in M-RoPE

- Index tokens so each frame's center sits at (t, t, t).
- Creates a smooth diagonal progression across text→video→text.
- Preserves consistent relative distances among visual tokens.

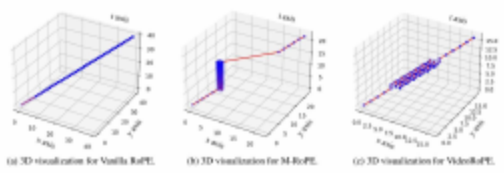


Figure: 3D位置编码对比: 1DRoPE无空间; M-RoPE跨帧索引不一致; VideoRoPE兼顾空间与一致索引增长

- Scale temporal indices by δ to decouple time vs. spatial/text granularity.
- Shift trailing text by $(\delta - 1) \cdot T_v$ to maintain continuity.
- Empirically, $\delta = 2$ works best on average.

- Long video understanding: LongVideoBench, MLVU, Video-MME (8s–2h).
- Retrieval: V-NIAH and new V-NIAH-D with periodic distractors.
- Hallucination: VideoHalluc; training on LLaVA-Video-178k subset (1.3M QA).



Figure: 五个视觉问答问题及其对应needle与distractor

Main Results Across Understanding Benchmarks

- VideoRoPE leads across LongVideoBench, MLVU, Video-MME (8k–64k).
- Notable gains: +2.91 (LVB 64k), +4.46 (MLVU 64k).
- Consistent best or second-best outside training range.

Table: Table 2. Comparison of different RoPE methods on LongVidionBench, MLVU, and Video-MME.

Method	LongVideoBench				MLVU				Video-MME			
	8k	16k	32k	64k	8k	16k	32k	64k	8k	16k	32k	64k
Vanilla RoPE (Su et al., 2024)	54.97	54.87	54.56	54.04	63.31	65.79	65.93	62.02	60.67	60.00	61.33	58.33
TAD-RoPE (Gao et al., 2024)	54.14	55.08	53.94	53.42	63.67	65.28	65.28	60.73	60.33	61.33	62.00	58.67
M-RoPE (Wang et al., 2024a)	53.42	52.80	53.11	54.35	60.41	60.68	61.56	61.10	60.67	59.67	61.00	59.67
VideoRoPE (Ours)	54.46	55.29	57.15	57.26	65.19	66.29	66.02	65.56	61.33	61.00	61.67	61.33

- Baselines: Vanilla RoPE, TAD-RoPE, M-RoPE (fair comparisons).
- Initialize from Qwen2-VL-7B (vision) and Qwen2-7B (LLM).
- Videos at 2 fps (≤ 128 frames); apply VideoRoPE in LLM attention.

V-NIAH-D: Stress-Testing Temporal Robustness

- Periodic distractors near target frames confuse high-frequency time encodings.
- VideoRoPE's low-frequency time resists distractor aliasing.
- Qualitative design matches the identified failure modes.
- Example: question asks for “the red mug tip-over” at 8:20; distractors show similar red mugs every 30s—VideoRoPE still retrieves the true event.



Figure: 为验证频域分配：基于VIAH构建含干扰的V-NIAH-D；VideoRoPE较M-RoPE检索更抗干扰

- Large margins on V-NIAH and V-NIAH-D over M-RoPE.
- Demonstrates superior long-range temporal discrimination.
- Validates principled low-frequency temporal design.

Table: Table 3. Performance comparison of different RoPEs on V-NIAH and V-NIAH-D.

Method	V-NIAH Acc.	V-NIAH-D Acc.
Vanilla RoPE (Su et al., 2024)	31.78	30.22
TAD-RoPE (Gao et al., 2024)	29.33	29.56
M-RoPE (Wang et al., 2024a)	78.67	74.67
VideoRoPE	91.11	87.11

- Higher success across haystack lengths and frame depths.
- Generalizes beyond training contexts with minimal degradation.
- At 128k, VideoRoPE drops only -1.62 ($57.26 \rightarrow 55.64$), the smallest degradation among baselines.
- Supports scalability claims of VideoRoPE design.

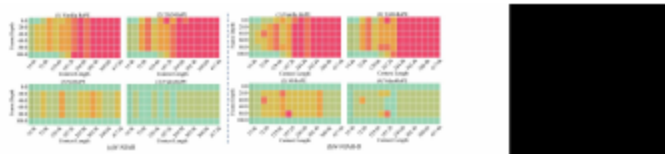


Figure: V-NIAH/V-NIAH-D检索可视化：绿→红示性能优→差

- M-RoPE over-relies on spatial cues; temporal attention remains local.
- VideoRoPE activates long-range temporal attention for retrieval.
- Explains correctness under distractors and long gaps.
- Example: a query frame at $t = 50$ attends to frames at $t \approx 10$ and $t \approx 90$ where the action recurs; M-RoPE attends locally instead.

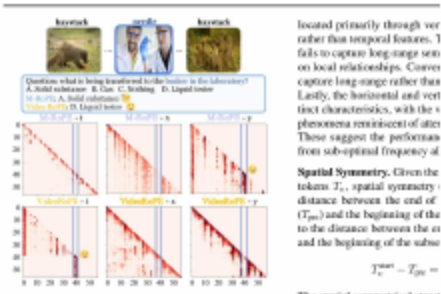


Figure: 注意力频率分配：M-RoPE仅局部对齐（对角），VideoRoPE沿时间检索针目标

- Temporal hallucination errors drop substantially (e.g., $+29.5\%$ absolute over baselines).
- Better object–relation grounding reduces cross-frame confusion.
- Example: when a person sets down an item off-screen, VideoRoPE avoids “still holding” claims.

- DL improves spatial symmetry; LTA reduces time oscillations; ATS aligns text/video granularity.
- Stacking DL+LTA+ATS yields the best 64k scores on LVB/MLVU.
- Example: with DL+LTA, retrieval remains stable when distractor frames repeat every 10–20s.
- $\delta = 2$ offers the best trade-off; too small underweights time; too large over-stretches.
- Interleaving x/y beats sequential allocation; [xyt] frequency allocation outperforms alternatives.

- Low-frequency temporal allocation helps even at 8k.
- More focused temporal attention improves QA reliability.
- Findings generalize across context sizes.



Figure: Additional visual analysis of attention.

Conclusion: What VideoRoPE Delivers

- Principled 3D RoPE with LTA, DL, ATS for long video reasoning.
- Better long-video retrieval, understanding, and reduced hallucination.
- Stable extrapolation up to 128k with minimal degradation.

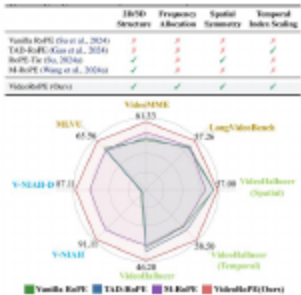


Figure: VideoRoPE outperforms RoPE variants on benchmarks.

Future Directions and Impact

- Learn δ jointly; adapt frequency per layer/head.
- Integrate with streaming/memory; extend to audio or 4D sensors.
- Synergize with hierarchical tokenization to push context further.

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- Yang, A., et al. 2024. Qwen2 Technical Report. <https://arxiv.org/abs/2407.10671>

- Thanks to collaborators, data providers, and open-source communities.
- We acknowledge benchmark maintainers and prior RoPE variants' authors.

- Thank you for your attention!
- Questions and feedback are welcome.